

POLYLOGARITHM VALUES AT A GOLDEN RATIO-BASED ARGUMENT

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ABSTRACT. This paper introduces and analyzes a specific complex constant, denoted Λ_{G1} , which is derived from inverse powers of the golden ratio, ϕ . We define the constant's components and prove that its magnitude is less than one. This validates the use of Λ_{G1} as an argument in the Polylogarithm function, $\text{Li}_s(z)$. We present high-precision numerical evaluations for the Dilogarithm ($s = 2$) and Trilogarithm ($s = 3$) of this constant. Based on this numerical evidence, we conjecture that the values $\text{Li}_s(\Lambda_{G1})$ are transcendental for all integers $s \geq 2$. Furthermore, we conjecture that these values do not belong to the field extension $\mathbb{Q}(\pi, \ln(2), \phi)$, a field chosen based on known polylogarithm identities. This line of inquiry is motivated in part by potential applications in the study of quasi-crystals.

1. INTRODUCTION

The study of special functions and constants often reveals connections between different areas of mathematics. This paper focuses on the analysis of the Polylogarithm function at a novel complex argument, Λ_{G1} , derived from the golden ratio, $\phi = (1 + \sqrt{5})/2$. A review of foundational literature in the field, such as Lewin's *Polylogarithms and Associated Functions* [3], reveals no prior study of this specific complex number as a polylogarithm argument. To substantiate this claim, a survey of recent literature and a search of major academic databases (including MathSciNet and the arXiv preprint server) were performed, which likewise showed no indication of this particular complex argument being studied, reinforcing the novelty of this investigation.

The motivation for this constant stems from the classical problem of dividing a line segment in extreme and mean ratio, as found in Book VI, Proposition 30 of Euclid's *Elements* [2]. We first rigorously define Λ_{G1} from inverse powers of ϕ and prove its validity as a polylogarithm argument. The analysis of the resulting values, $\text{Li}_s(\Lambda_{G1})$, forms the central topic of this work. Understanding the nature of these specific values may offer insights into mathematical domains where the golden ratio is foundational. For instance, given that the diffraction patterns of quasi-crystals are intrinsically linked to the golden ratio and that polylogarithms frequently arise in quantum field theory calculations (often in the context of Feynman diagrams), this work explores the intriguing possibility that values like $\text{Li}_s(\Lambda_{G1})$ might emerge as fundamental components in the theoretical description of such aperiodic structures. The paper culminates in a formal conjecture about the transcendental nature of these values, based on high-precision numerical analysis.

2. THE CONSTANT Λ_{G1} AND ITS FOUNDATIONAL PROPERTIES

2.1. Defining Components from the Golden Ratio. We define two real constants, T and J , directly from the algebraic properties of the golden ratio.

Definition 2.1 (Primary Real Components). The constants T and J are defined from the golden ratio, ϕ , as:

$$T = \frac{1}{2\phi} \quad \text{and} \quad J = \frac{1}{2\phi^2}$$

These definitions lead to a pair of linear relations that highlight their connection to classical geometry; as will be further elaborated in Appendix A, they are chosen to correspond to the division of a line segment in the golden ratio.

Proposition 2.2 (System of Equations). *The constants T and J are the unique real numbers (a, b) that satisfy the system:*

$$\begin{aligned} (1) \quad & a/b = \phi \\ (2) \quad & a + b = 1/2 \end{aligned}$$

Proof. The ratio is immediate from the definitions: $T/J = (1/(2\phi))/(1/(2\phi^2)) = \phi^2/\phi = \phi$. For the sum, we use the identity $\phi + 1 = \phi^2$:

$$T + J = \frac{1}{2\phi} + \frac{1}{2\phi^2} = \frac{\phi + 1}{2\phi^2} = \frac{\phi^2}{2\phi^2} = \frac{1}{2}$$

The two equations $a/b = \phi$ (or $a - \phi b = 0$) and $a + b = 1/2$ form a linear system in variables (a, b) . The determinant of the coefficient matrix is $\begin{vmatrix} 1 & -\phi \\ 1 & 1 \end{vmatrix} = 1 - (-\phi) = 1 + \phi$. Since $1 + \phi \neq 0$, the system has a unique solution. $\square$

Remark. This formulation shows that the constraint $T + J = 1/2$ is not an arbitrary choice, but a direct consequence of defining the components from successive inverse powers of ϕ . Using $\phi = (1 + \sqrt{5})/2$, we can find the explicit values. For T :

$$T = \frac{1}{2\phi} = \frac{1}{1 + \sqrt{5}} = \frac{1}{1 + \sqrt{5}} \cdot \frac{\sqrt{5} - 1}{\sqrt{5} - 1} = \frac{\sqrt{5} - 1}{5 - 1} = \frac{\sqrt{5} - 1}{4}$$

For J , using $\phi^2 = \phi + 1 = (3 + \sqrt{5})/2$:

$$J = \frac{1}{2\phi^2} = \frac{1}{3 + \sqrt{5}} = \frac{1}{3 + \sqrt{5}} \cdot \frac{3 - \sqrt{5}}{3 - \sqrt{5}} = \frac{3 - \sqrt{5}}{9 - 5} = \frac{3 - \sqrt{5}}{4}$$

Thus, the explicit values are $T = (\sqrt{5} - 1)/4$ and $J = (3 - \sqrt{5})/4$.

From these constants, we define our primary object of study.

Definition 2.3 (The Primary Constant). The constant Λ_{G1} is defined as the complex number $\Lambda_{G1} = T + iJ$.

2.2. Convergence Property. A crucial property of Λ_{G1} is that its magnitude is less than one, which is required for the convergence of the Polylogarithm series.

Lemma 2.4 (Magnitude of Λ_{G1}). *The constant Λ_{G1} lies within the unit disk, i.e., $|\Lambda_{G1}| < 1$.*

Proof. We require $|\Lambda_{G1}|^2 < 1$. We calculate the squared values of the components: $T^2 = \left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{6-2\sqrt{5}}{16}$ and $J^2 = \left(\frac{3-\sqrt{5}}{4}\right)^2 = \frac{14-6\sqrt{5}}{16}$. Summing these gives:

$$\begin{aligned} |\Lambda_{G1}|^2 &= T^2 + J^2 = \frac{6 - 2\sqrt{5} + 14 - 6\sqrt{5}}{16} \\ &= \frac{20 - 8\sqrt{5}}{16} = \frac{5 - 2\sqrt{5}}{4} \approx 0.131966 \end{aligned}$$

Since $0.131966 < 1$, the condition is satisfied. $\square$

3. EVALUATING THE POLYLOGARITHM AT Λ_{G1}

The property $|\Lambda_{G1}| < 1$ is central to this paper, as it allows us to define the Polylogarithm values at this constant. The Polylogarithm function is a key object in this study.

Definition 3.1 (The Polylogarithm Function). For any complex s , the Polylogarithm function of order s , $\text{Li}_s(z)$, is defined by the power series:

$$\text{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}$$

This series converges for all complex numbers z such that $|z| < 1$.

Since Λ_{G1} satisfies the convergence condition, we can define the specific numerical values that are the subject of our main conjecture:

$$\text{Li}_s(\Lambda_{G1}) = \sum_{n=1}^{\infty} \frac{(\Lambda_{G1})^n}{n^s}$$

This identification holds for any complex s .

4. NUMERICAL ANALYSIS AND A CONJECTURE ON THE NATURE OF $\text{Li}_s(\Lambda_{G1})$

4.1. High-Precision Numerical Evaluations. We present high-precision numerical evaluations for the first two non-trivial integer cases of the Polylogarithm of Λ_{G1} . The values were computed using the Python `mpmath` library with a working precision of 200 decimal digits to ensure robustness.

- For $s = 2$, the Dilogarithm of Λ_{G1} is:

$$\text{Li}_2(\Lambda_{G1}) \approx 0.322328253720993 + 0.226730769307749i$$

- For $s = 3$, the Trilogarithm of Λ_{G1} is:

$$\text{Li}_3(\Lambda_{G1}) \approx 0.310345112613931 + 0.198884393433621i$$

4.2. A Conjecture on the Nature of These Values. The transcendental nature of special function values at algebraic arguments is a central topic in number theory [1]. For context, values of the Dilogarithm are known to be transcendental at certain related real algebraic points, such as $\text{Li}_2(1/2) = \frac{\pi^2}{12} - \frac{1}{2} \ln^2(2)$ and $\text{Li}_2(\phi^{-2}) = \frac{\pi^2}{15} - \ln^2(\phi)$. The constant Λ_{G1} , being a non-real algebraic number that is not a root of unity, presents a distinct and challenging case. Transcendence proofs for polylogarithm values frequently rely on techniques that do not readily extend to a general complex algebraic argument like Λ_{G1} . Many powerful results in this area are restricted to arguments that are rational or real, while proofs for complex arguments often require them to be roots of unity. The constant Λ_{G1} falls into none of these categories, placing its analysis outside the scope of many established methods.

To investigate the nature of these values, we employed the PSLQ integer relation algorithm, implemented in Python's `mpmath` library [4]. For both the Dilogarithm ($s = 2$) and Trilogarithm ($s = 3$) values, the real and imaginary components were tested separately for a null linear combination against a basis of real constants. This basis consisted of elements of the form $\pi^a (\ln 2)^b \phi^c$, where a, b, c are non-negative integers such that the total degree $a + b + c < 5$. This basis was chosen because its elements are constructed from the kinds of constants that appear in known closed-form identities for polylogarithms at related algebraic arguments, such as the well-known identity $\text{Li}_2(\phi^{-2}) = \frac{\pi^2}{15} - \ln^2(\phi)$, making it a plausible test set for discovering simple dependencies. No relation was found with a detection tolerance of 10^{-190} . While not a formal proof, this failure provides strong numerical motivation for their apparent transcendental nature and leads us to propose the following general conjecture.

Conjecture 4.1. *For any integer $s \geq 2$, the value $\text{Li}_s(\Lambda_{G1})$ is a transcendental number. Furthermore, it is conjectured that this number is not in the field $\mathbb{Q}(\pi, \ln(2), \phi)$.*

Remark. Proving this conjecture is beyond the scope of this paper, but proposing it frames a clear direction for future work. The field in the second part of the conjecture is chosen to reflect the basis of constants used in our numerical search; a wider search including other fundamental constants (such as Euler's constant γ) may be a fruitful avenue for future analysis, as noted in the Future Work section.

5. CONCLUSION

This paper has introduced a complex constant, Λ_{G1} , whose definition is derived non-arbitrarily from the golden ratio. We have proven that its magnitude is less than one, which validates its use as an argument in the Polylogarithm function. The primary contribution of this work is the introduction of this constant and the subsequent analysis of its polylogarithmic values, $\text{Li}_s(\Lambda_{G1})$. Based on high-precision numerical evidence from the $s = 2$ and $s = 3$ cases, we have proposed a new open problem: a conjecture on the transcendental nature of these values.

6. FUTURE WORK

The conjecture presented in Section 4 opens a clear path for future investigation. A primary goal remains to prove or disprove the transcendental nature of $\text{Li}_s(\Lambda_{G1})$. More specific avenues for future work include:

- Separately analyzing the real and imaginary parts of $\text{Li}_s(\Lambda_{G1})$ and determining if they are transcendental or satisfy relations with other known constants.
- Expanding the numerical search for relations by using a larger basis of constants (e.g., including Euler’s constant γ , Catalan’s constant) to test for more complex identities, alongside an increased search depth.
- Investigating further algebraic properties of Λ_{G1} , such as its minimal polynomial or its potential role in other integral or series evaluations.
- Investigating potential applications in physics, particularly in the study of quasi-crystals. Given the link between the golden ratio and diffraction patterns, and the appearance of polylogarithms in quantum field theory calculations, these values may be fundamental in describing aperiodic structures.
- Exploring related series identities, such as the evaluation using Eulerian polynomials shown in Appendix B, to see if they reveal further structure or properties of $\text{Li}_s(\Lambda_{G1})$.

APPENDIX A. GEOMETRIC MOTIVATION FOR THE DEFINING EQUATIONS

The definitions of T and J have a direct geometric motivation arising from the division of a line segment according to the golden ratio. If a segment of length L is divided into two pieces, a and b , such that $a/b = \phi$, the pieces have lengths $a = L/\phi$ and $b = L/\phi^2$. Our constants T and J correspond to this construction for $L = 1/2$. As shown in Proposition 2.2, the sum $T + J = 1/2$ arises as a direct consequence of defining the components from the fundamental quantities $1/(2\phi)$ and $1/(2\phi^2)$. This provides a non-arbitrary basis for the constant’s definition.

APPENDIX B. EVALUATION OF A RELATED POLYNOMIAL-WEIGHTED SERIES

As a consequence of Lemma 2.4, Λ_{G1} is a valid argument in the series whose sum is given by Eulerian polynomials, $A_k(x)$. This provides a direct evaluation for a family of series with Λ_{G1} as the base. For any integer $k \geq 0$:

$$\sum_{n=0}^{\infty} n^k \cdot (\Lambda_{G1})^n = \frac{A_k(\Lambda_{G1})}{(1 - \Lambda_{G1})^{k+1}}$$

While not central to the paper’s main conjecture, this identity is an interesting property of the constant and demonstrates its utility in evaluating other classes of well-known series.

REFERENCES

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